# One-way Analysis of Variance (ANOVA)

## *Within-subjects designs*

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# Έλεγχος υποθέσεων-Βήματα

1. Καθορίζεται η **μηδενική υπόθεση H0 ( = )** και **εναλλακτική υπόθεση H1 ( ≠ ) .**

2. Ορίζεται το **επίπεδο σημαντικότητας** α (συνήθως α=0.05).

3. Επιλέγεται μια κατάλληλη **στατιστική δοκιμασία** και υπολογίζεται η τιμή του στατιστικού με βάση τα δεδομένα του δείγματος.

4. Σύγκριση της **πιθανότητας p** να έχουμε την συγκεκριμένη τιμή του στατιστικού (ή κάτι πιο ακραίο) θεωρώντας ότι ισχύει η Ho, με το **επίπεδο σημαντικότητας α** (0.05). Στατιστικά σημαντικό αποτέλεσμα ($p < 0.05$).

5. **Ερμηνεία** αποτελεσμάτων.

# One-way within-subjects ANOVA

ANOVA: Y = numeric variable → **Dependent**

X = categorical (factor) variable (>2 levels) → **Independent**

Each subject is exposed to all levels of the independent variable.

**NOTE:** If categorical variable has only 2 related samples: paired t-test

EXAMPLE

A study investigated the effect of a drug on symptom intensity in a specific disease, assessing eight patients before (two pre-test baseline measures) and after treatment (three post-test measures). The primary question is whether the mean symptom intensity score changed over time.

Y: Symptom intensity in a specific disease (0-12)
X: Follow-up time (five time points: 1 month and 1 week before treatment, 1 week, 1 month, 1 year after treatment)

# One-way within-subjects ANOVA: hypotheses

$H_0$: The means of all the **related** groups are equal ($\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$).

$H_1$: At least one mean differs from the other

- doesn't say how or which ones differ.
- Can follow up with "post-hoc tests"

# Assumptions of one-way within-subjects ANOVA

- **Within-subjects design** (each participant is exposed to all levels of the independent variable)

- **Normality:** The outcome variable (symptom intensity in our example) must be normally distributed in each group (here five time points: 2 baseline and 3 after treatment measurements).

  Check this by looking at histograms, Q-Q plots, and Shapiro-Wilk test:

  If $P>0.05$ then normal distribution

  If $P<0.05$ then non-normal distribution

- **Sphericity assumption:** The variances of the differences between all possible pairs of within-subject conditions are equal.

  Check sphericity with the Mauchly's sphericity test

  If $P>0.05$ then ANOVA

  If $P<0.05$ then ANOVA with corrected df (Greenhouse-Geisser (GGe), or Huynh-Feldt (HFe))

# DATA

| Factor levels | time point | | | | | Sum |
|---|---|---|---|---|---|---|
| | 1 month before | 1 week before | 1 week afterr | 1 month after | 1 year after | |
| S1 | 12 | [illegible] | [illegible] | 5 | 5 | 41 |
| S2 | 9 | [illegible] | [illegible] | 6 | 5 | 35 |
| S3 | 9 | [illegible] | [illegible] | 5 | 6 | 35 |
| S4 | 8 | [illegible] | [illegible] | 4 | 4 | 28 |
| S5 | 8 | [illegible] | [illegible] | 4 | 5 | 32 |
| S6 | 9 | 10 | 8 | 6 | 7 | 40 |
| S7 | 12 | 10 | 7 | 5 | 4 | 38 |
| S8 | 6 | 7 | 5 | 7 | 5 | 30 |
| **Sum** | 73 | 72 | 51 | 42 | 41 | |
| **Mean** | 9.13 | 9.00 | 6.38 | 5.25 | 5.13 | |

symptom intensity scores

# DATA

| Factor levels | 1 month before | 1 week before | 1 week afterr | 1 month after | 1 year after | Sum |
|---|---|---|---|---|---|---|
| S1 | 12 | 12 | 7 | 5 | 5 | 41 |
| S2 | 9 | 10 | 5 | 6 | 5 | 35 |
| S3 | 9 | 8 | 7 | 5 | 6 | 35 |
| S4 | 8 | 6 | 6 | 4 | 4 | 28 |
| S5 | 8 | 9 | 6 | 4 | 5 | 32 |
| S6 | 9 | 10 | 8 | 6 | 7 | 40 |
| S7 | 12 | 10 | 7 | 5 | 4 | 38 |
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Between subjects (residuals)

# DATA

| Factor levels | 1 month before | 1 week before | 1 week afterr | 1 month after | 1 year after | Sum |
|---|---|---|---|---|---|---|
| S1 | 12 | 12 | 7 | 5 | 5 | 41 |
| S2 | 9 | 10 | 5 | 6 | 5 | 35 |
| S3 | 9 | 8 | 7 | 5 | 6 | 35 |
| S4 | 8 | 6 | 6 | 4 | 4 | 28 |
| S5 | 8 | 9 | 6 | 4 | 5 | 32 |
| S6 | 9 | 10 | 8 | 6 | 7 | 40 |
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| **Mean** | 9.13 | 9.00 | 6.38 | 5.25 | 5.13 | |

Within subjects: time

Treatment effect

## DATA

| Factor levels | 1 month before | 1 week before | 1 week after | 1 month after | 1 year after | Sum |
|---|---|---|---|---|---|---|
| S1 | 12 | 12 | 7 | 5 | 5 | 41 |
| S2 | 9 | 10 | 5 | 6 | 5 | 35 |
| S3 | 9 | 8 | 7 | 5 | 6 | 35 |
| S4 | 8 | 6 | 6 | 4 | 4 | 28 |
| S5 | 8 | 9 | 6 | 4 | 5 | 32 |
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| S7 | 12 | 10 | 7 | 5 | 4 | 38 |
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| **Sum** | 73 | 72 | 51 | 42 | 41 | |
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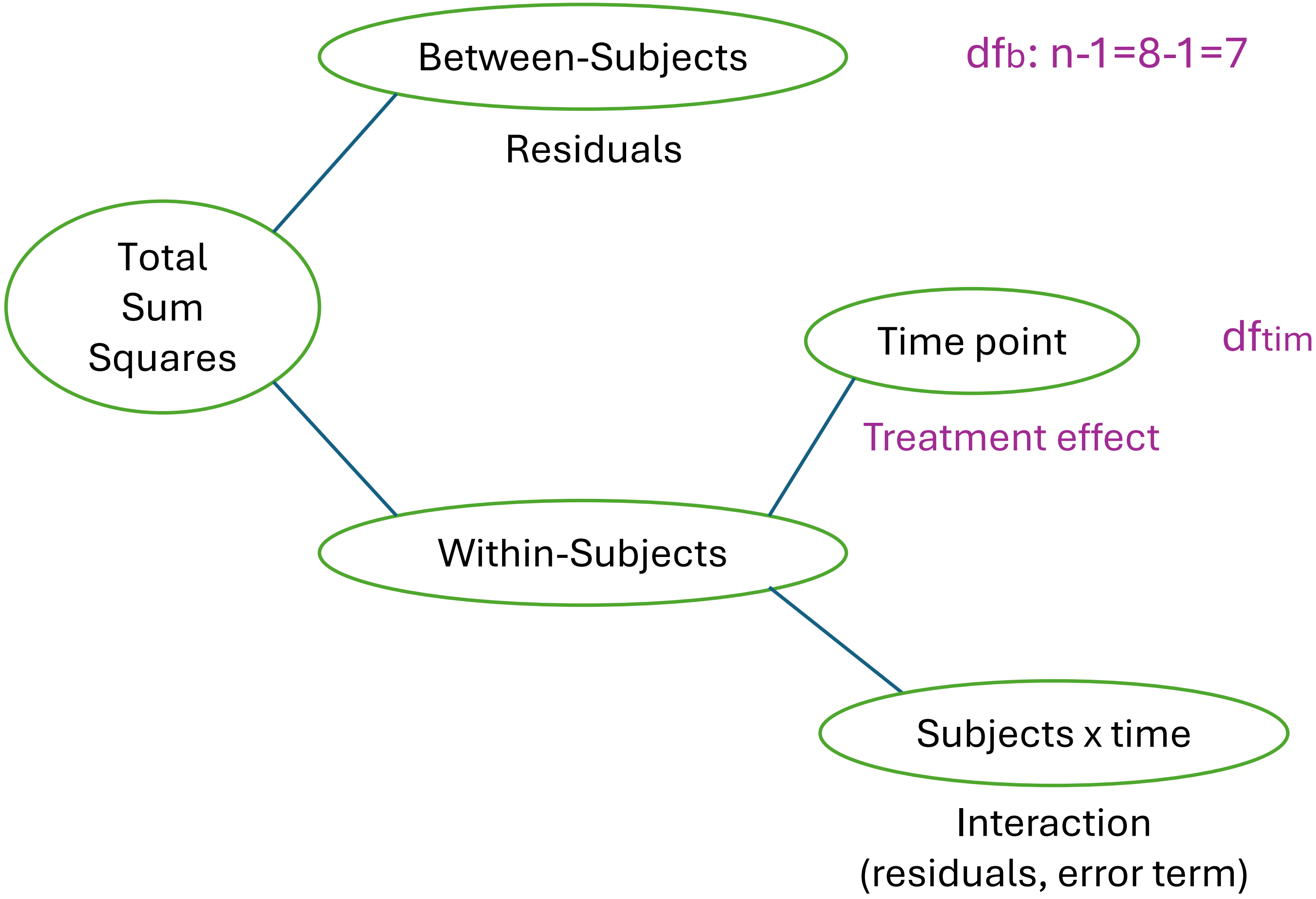
Within subjects: subjects x time

Interaction
(residuals, error term)

**One-way within- subjects ANOVA**

## Partitioning the total variance into its sources




$df_b$: n-1=8-1=7

$df_{timepoint}$: k-1=5-1=4

$Df_{subject \times timepoint}$:
(n-1) x (k-1) = 7 x 4= 28

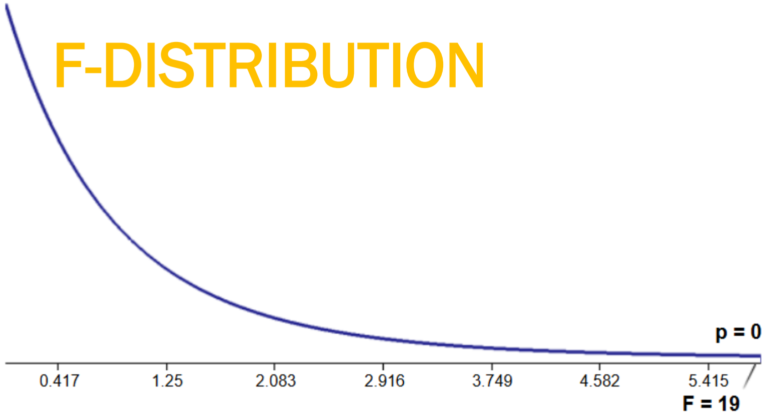
Within Subjects Effects

| | Sphericity Correction | Sum of Squares | df | Mean Square | F | p |
|---|---|---|---|---|---|---|
| timepoint | Greenhouse-Geisser | 123.85000 | 1.950930 | 63.482554 | 18.62406 | 0.0001340 |
| Residual | Greenhouse-Geisser | 46.55000 | 13.656508 | 3.408631 | | |

Note. Type 3 Sums of Squares

***NOTE:*** The degrees of freedom are corrected using the GG epsilon

F-DISTRIBUTION




Between Subjects Effects

| | Sum of Squares | df | Mean Square | F | p |
|---|---|---|---|---|---|
| Residual | 30.57500 | 7 | 4.367857 | | |

Note. Type 3 Sums of Squares

# Post hoc Tests

- Bonferroni Procedure
- Tukey’s Test
- Scheffe
- Holm